Timothy James Wood

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SUMMARY

Research specialising in statistical population genetics, with hands-on experience in demographic inference and genome-wide selection scans under the guidance of leading academics. Currently pursuing an MSc in Bioinformatics, bringing a solid foundation in machine learning and statistical methodologies applied to genomic data. Proven ability to analyse complex datasets and derive meaningful insights, seeking to leverage advanced research skills in a PhD programme.

EDUCATION

University of Bristol Sept 2026 – Sept 2027
MSc Bioinformatics *TBC*

- HDR UK MSc Scholarship (charity partner: Coeliac UK)

University of Bristol Sept 2023 – Jun 2026
BSc Biology *First Class Honours*

- First-class marks in both the final-year research project (75%) and literature review (78%)

TECHNICAL SKILLS

Programming	Python (NumPy, SciPy, pandas), R (tidyverse), Bash, Git, SLURM/HPC, LaTeX
Statistical genetics	dadi, pcadapt, VCFtools, PLINK; composite likelihood, block bootstrap, PCA
Laboratory	PCR / qPCR, gel electrophoresis, DNA extraction, Sanger sequencing, mass spectrometry

RESEARCH EXPERIENCE

School of Mathematics, University of Bristol Jun – Sept 2026
Research Assistant for Prof. Mark Beaumont and Dr Dennis Prangle *timj-wood/wildcat_models*

- Fitted three demographic models in dadi to a joint SFS from 22 wildcat and domestic cat genomes
- Implemented composite-likelihood model selection and Godambe confidence intervals from first principles
- Found a boundary optimum invalidating the top-scoring model; showed deep parameters lie on a likelihood ridge
- Ran optimisation as SLURM array jobs on Bristol's HPC; results form the baseline for simulation-based inference

BSc Research Project: Selection and Introgression in the European Wildcat Sept – Dec 2025
Supervisor: Prof. Mark Beaumont *Data from Dr Jo Howard-McCombe, RZSS*

- Processed whole-genome SNP data for 108 wild, captive and domestic cats (VCFtools, PLINK)
- Ran a pcadapt genome scan for selection; identified 1,275 outlier intervals and annotated 11 candidate genes
- Found significant GO enrichment for GABA, taurine and calcium-ion transport
- Developed PCA visualisations to separate introgression from demographic structure

BSc Literature Review: Engineering Predictability in Living Systems Sept – Dec 2025
Supervisor: Prof. Thomas Gorochowski *Biocompute Lab*

- Reviewed synthetic circuits, genome recoding and biological foundation models (Evo, ESM-2, ProGen)
- Argued that data-driven models capture evolutionary correlation rather than causal mechanism
- Proposed context-aware data and adaptive feedback as remedies

OTHER EXPERIENCE

PwC LLP Jun – Jul 2025
Audit Intern *London*

- Facilitated the audit of WPP plc (FTSE 100); produced analytical working papers from large financial datasets
- Trained in IFRS and audit methodology; produced review-ready working papers and schedules in Excel
- Secured a graduate return offer; turned down to pursue further study